



Medical Image Analysis 8BE030

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Department of Biomedical Engineering, Medical Image Analysis group

Outline for today:

- Course introduction
- Introduction to image registration
 - Causes of misalignment
 - Applications of medical image registration
 - Classification of image registration methods
- Geometrical transforms
 - Recap linear algebra
 - Rigid and affine transformations
 - Non-linear transformations

Lecturers



Dr. Ruisheng Su – r.su@tue.nl

Background: BSc in Electrical Engineering (Shandong University, China), MSc in Electrical Engineering (Technical University of Munich, Germany), PhD in Medical Image Analysis (Erasmus MC, Erasmus University Rotterdam)

Research: Deep learning, medical image analysis in neurovascular diseases, image-guided interventions



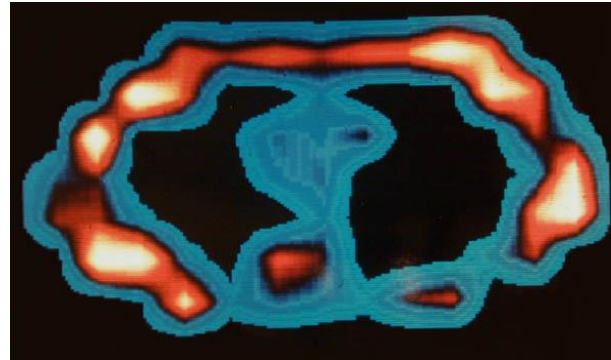
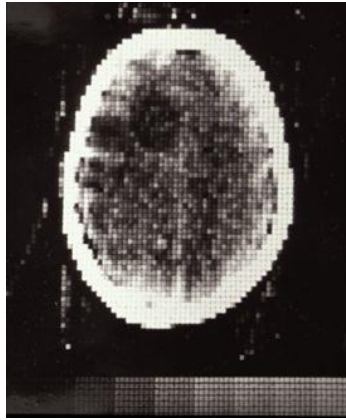
Dr. Cian Scannell – c.m.scannell@tue.nl

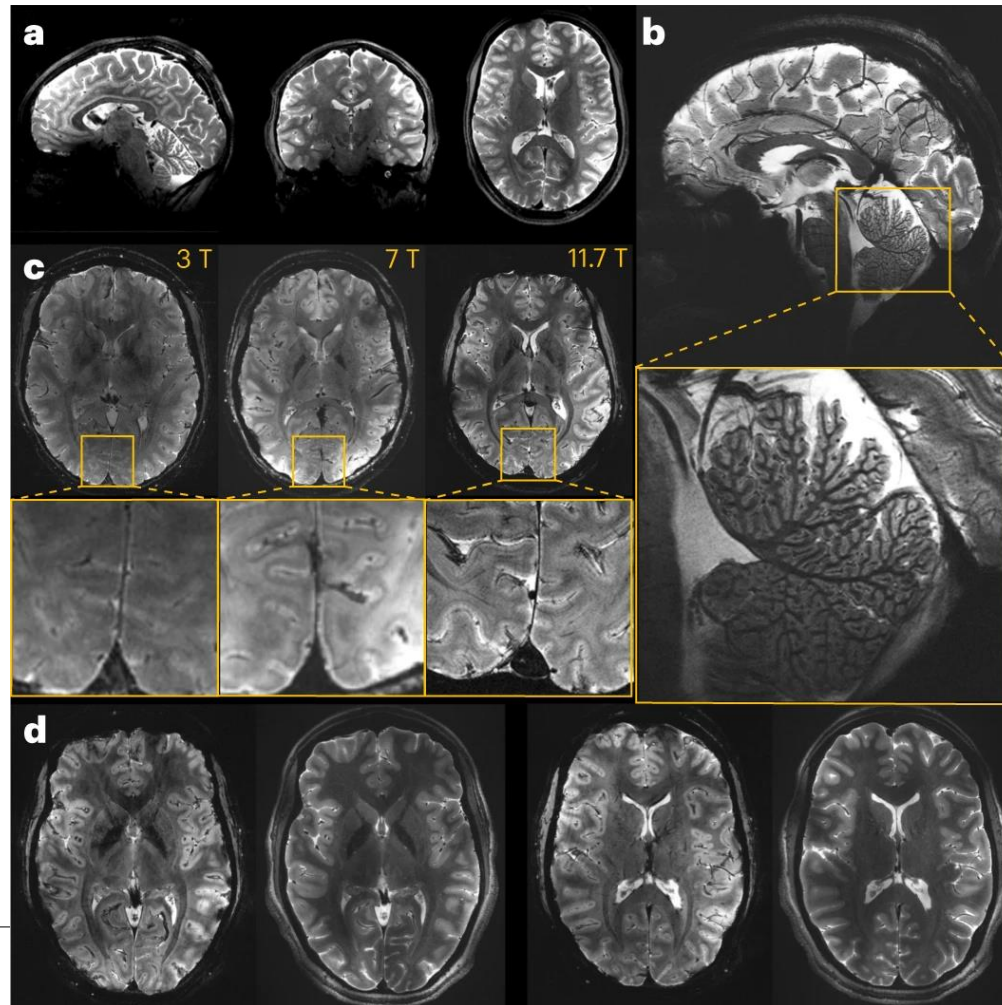
Background: BSc in Mathematical Sciences (University College Cork, Ireland), MRes in Medical Imaging, PhD in Biomedical Engineering (King's College London, UK)

Research: Deep learning, quantitative MRI, cardiovascular imaging & modelling

Student teaching assistants:

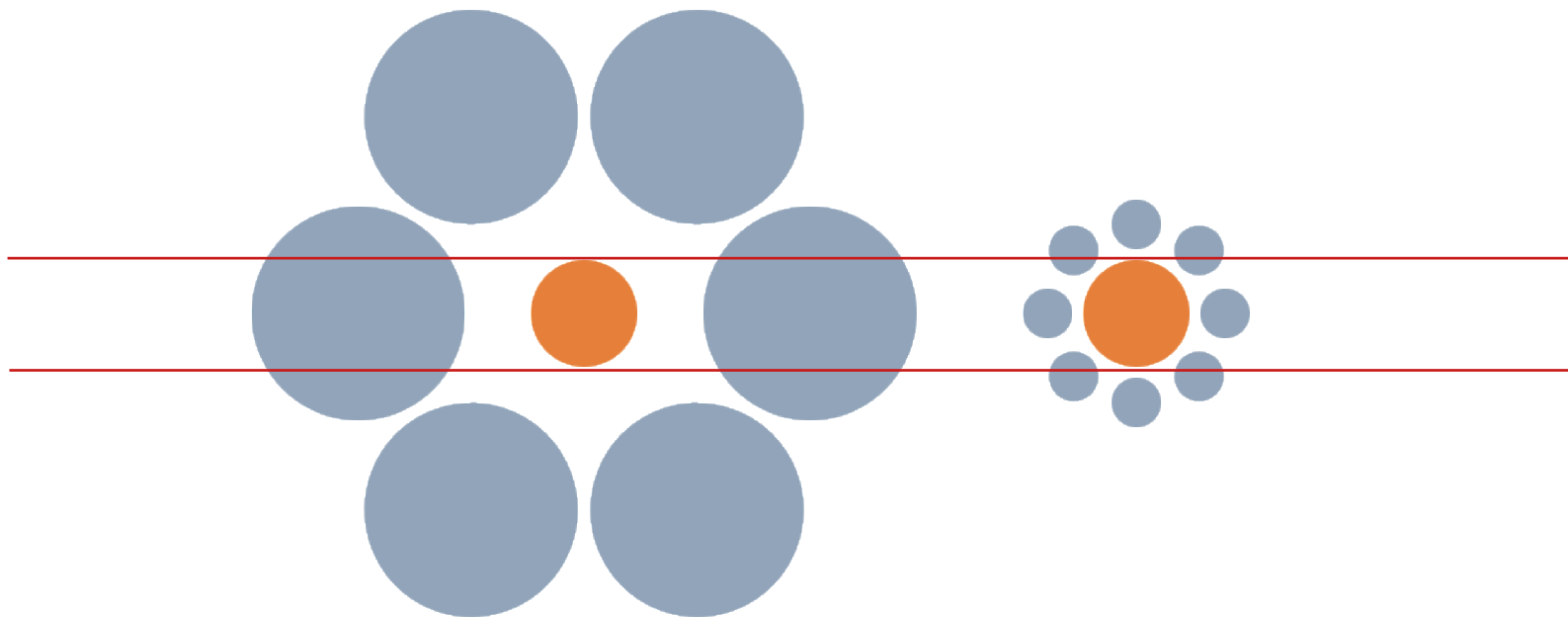
- Simon Habraken
- Joris Mentink
- Silke Peters
- Dirk Sollewijn Gelpke

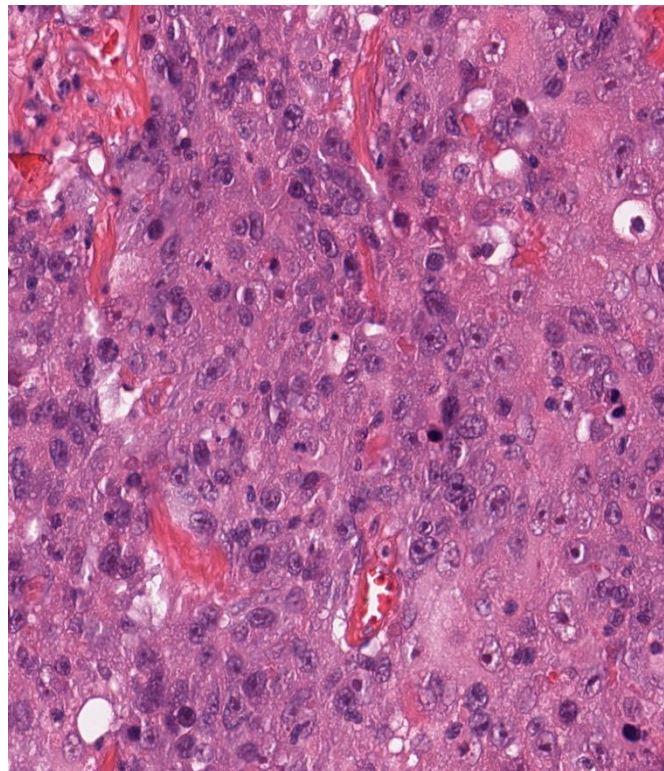
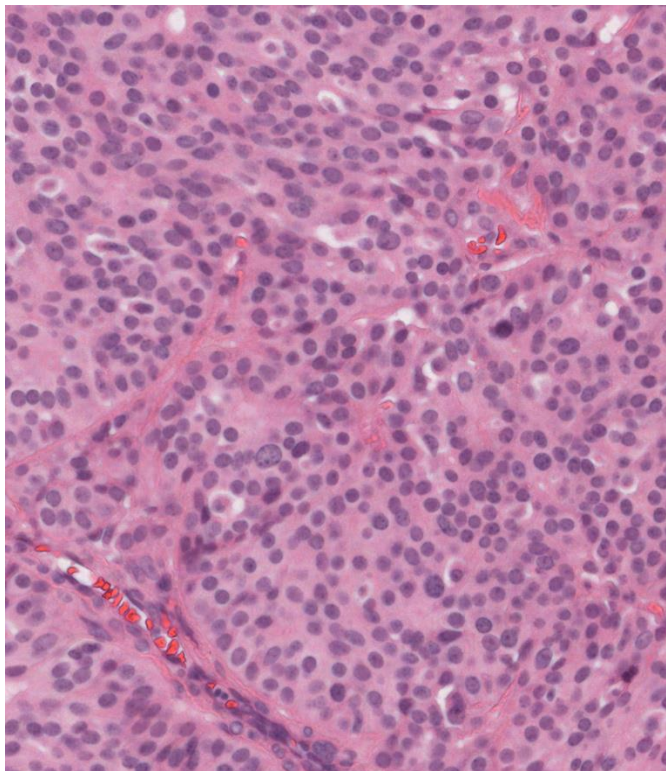


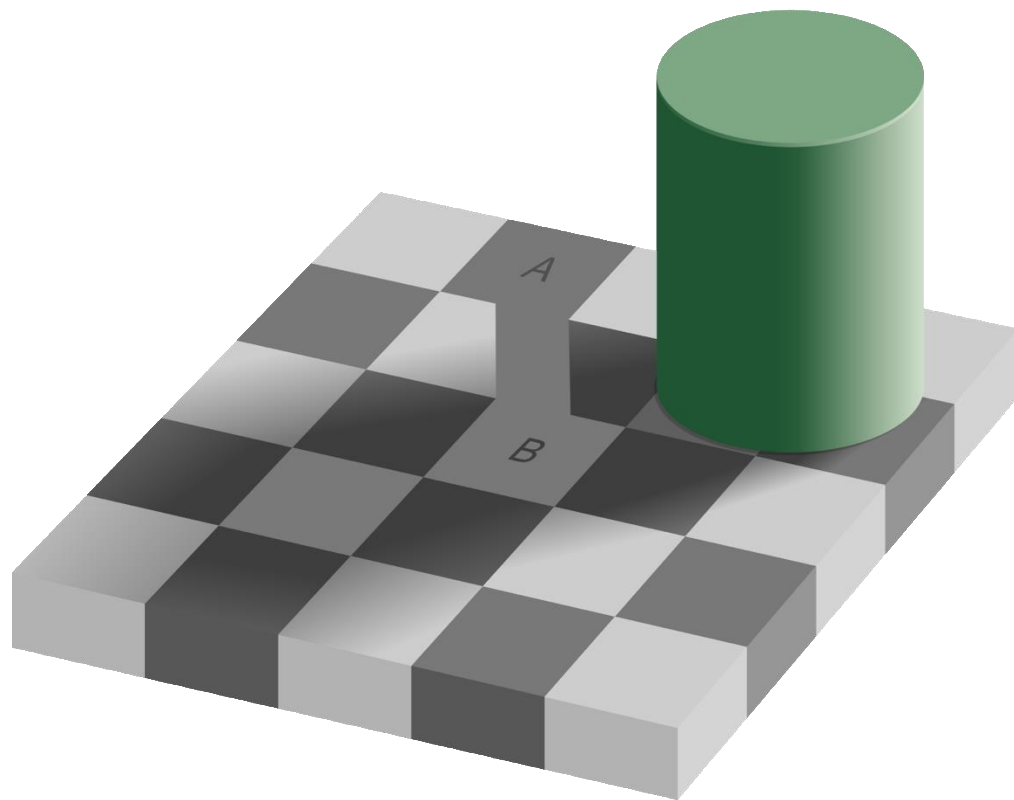


In this course, we'll explore how advanced image analysis techniques can help tackle real clinical problems.

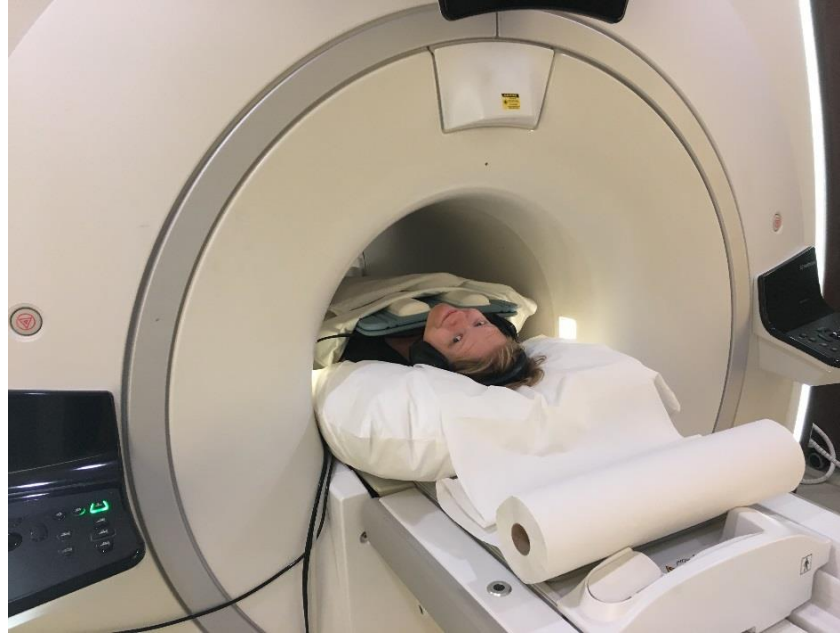
But why automatic image analysis?







Present day in the Netherlands: 9K CT scans, 5K MR scans per 100K people per year



Understanding and Appreciating Burnout in Radiologists

 Christopher R. Bailey¹, Allison M. Bailey¹, Anna Sophia McKenney,  Clifford R. Weiss 



Contents lists available at [ScienceDirect](https://www.sciencedirect.com)

European Journal of Radiology Open

journal homepage: www.elsevier.com/locate/ejro



Incidence and factors associated with burnout in radiologists: A systematic review

Nader A. Fawzy^a, Muhammad
Tamara Alsheikh^a, Ali Ahmed

RADIOLOGICAL EDUCATION

Prevalence of burnout among German radiologists: a call to action

Moritz B. Bastian^{1*} , Laureen Fröhlich¹, Joel Wessendorf¹, Michael Scheschenja¹, Alexander M. König¹, Jarmila Jedelska¹ and Andreas H. Mahnken¹



The goal of medical image analysis is to develop **automated** methods that enable **faster, more reliable** and **quantitative** analysis of medical images.

Learning goals

After completing the course, the student...

..has insight of the role of medical image analysis tasks in addressing clinical questions.

... has knowledge of how basic engineering and mathematical techniques can be used to design medical image analysis methods.

... can implement and apply medical image analysis methods.

... can analyze the results of medical image analysis methods.

Place of the course in the curriculum:

8BB050 Image Acquisition and Processing

8BE030 Medical Image Analysis

8P361 Project AI Imaging (3rd year BSc)

BEP in Medical Image Analysis

Master:

8DM20 Capita Selecta in Medical Image Analysis

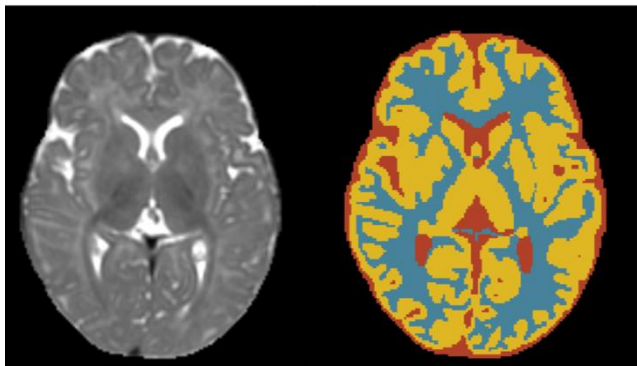
8DM40 Machine Learning in Medical Imaging and Computational Biology

Graduation project and internship

Overview of different medical image analysis tasks (2D, 3D, 3D+, ...)

This course

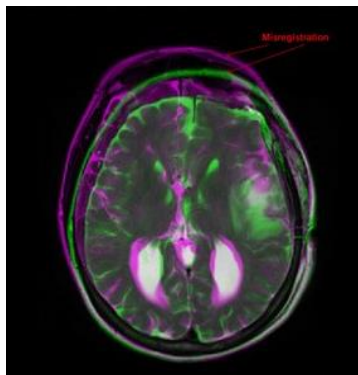
Image segmentation



Dividing an image into multiple regions with similar properties (e.g., intensity values).

NB: these regions typically correspond to different anatomical structures.

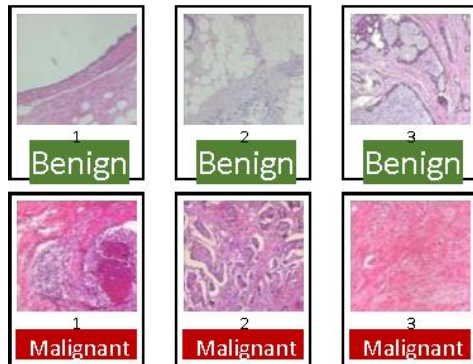
Image registration



Finding an optimal transformation that aligns two images.

8BB020

Computer-aided detection (CAD)



Categorizing/labeling images based on specific rules.

Official definition: “systems that assist doctors in the interpretation of medical images, often based on machine learning”

Course organization

Two main topics:

1. Medical image registration (Ruisheng)
2. Medical image segmentation (Ruisheng + Cian)
3. Deep learning for registration & segmentation (Cian + Ruisheng)

Lectures, exercises & project work

Course schedule

Monday and Thursday: lectures and practicals (guided self-study)

Practicals:

You can work in **groups of up to 4 students** on:

- Exercises
- Project work

You can sign yourself up into a group on Canvas.

Report must contain a paragraph describing the contributions of each group member which can be used to to adjust individual grades to reflect a student's (lack of) contribution to the group

Exercises

Goals:

- Help you study the material
- Develop code that can be used for the project work
- Prepare for exam
- Not graded

Projects

- 2 projects (registration, segmentation)
- Each project is based on one or more **research questions** that you formulate yourself
- Short report & code
- Graded

Detailed description of the project deliverables and assessment rubric can be found in the project handouts.

Guided project work: questions and extension of the code developed in the exercises that will guide you to a **minimal project solution**.

Assessment

Projects

- Medical image registration (15%)
- Medical image segmentation (15%)

Written exam (70%), out of which some of the questions related to the project, i.e. project accounts for ~50% of the grade

Communication – digital platforms we will use during this course



- Communication
- Hand in assignments (project reports & code)
- Handouts lectures



- Course overview
- Python code for exercises and projects
- Handouts lectures

Communication

Main communication channel is Canvas: post your questions in the Discussion section.

Emailing is only for individual circumstances and not related to content. If you do email, use the tag [8BE030] in the subject line.

How to effectively ask questions?

- **Start the question by explaining the context**
 - State the goal of the task you are working
- **Formulate a specific question**
 - “I don’t know how to solve Exercise 2” is not a specific question.
 - Be clear and honest about what you want to get out.
 - “Is this enough for the project work?” is not allowed.

How to effectively ask questions?

- **Demonstrate that you have attempted to answer the questions or solve the problem**
 - Formulate a provisional answer (does not matter if it is correct or not)
- **Python:**
 - Read the documentation
 - Error messages are informative!
 - Before asking for help, make sure that your problem is reproducible

How to get started with the exercises and project work?

GitHub page: <https://github.com/tueimage/8be030-mia>

Follow software installation instructions:

- Anaconda / packages
- Python
- Jupyter

If you prefer a GUI: Anaconda Navigator

The screenshot displays the Anaconda Navigator application window. The interface includes a top menu bar with 'File' and 'Help', a sidebar on the left with navigation options like 'Home', 'Environments', 'Learning', and 'Community', and a main panel showing a grid of application tiles. The tiles are organized by environment, currently set to 'media'. The applications shown are:

- CMD.exe Prompt** (0.1.1): Run a cmd.exe terminal with your current environment from Navigator activated. [Launch]
- Jupyter Notebook** (6.0.3): Web-based, interactive computing notebook environment. Edit and run human-readable docs while describing the data analysis. [Launch]
- Powershell Prompt** (0.0.1): Run a Powershell terminal with your current environment from Navigator activated. [Launch]
- Qt Console** (4.7.5): PyQt GUI that supports inline figures, proper multiline editing with syntax highlighting, graphical calltips, and more. [Launch]
- Spyder** (4.1.3): Scientific Python Development Environment. Powerful Python IDE with advanced editing, interactive testing, debugging and introspection features. [Launch]
- Glueviz** (0.15.2): Multidimensional data visualization across files. Explore relationships within and among related datasets. [Install]
- JupyterLab** (2.1.5): An extensible environment for interactive and reproducible computing, based on the Jupyter Notebook and Architecture. [Install]
- Orange 3** (3.23.1): Component based data mining framework. Data visualization and data analysis for novice and expert. Interactive workflows with a large toolbox. [Install]
- RStudio** (1.1.456): A set of integrated tools designed to help you be more productive with R. Includes R essentials and notebooks. [Install]

Annotations with red arrows point to the 'Start a prompt (terminal)' text pointing to the CMD.exe Prompt tile, and 'Spyder' pointing to the Spyder tile. The text 'Jupyter Notebook' is also present in the lower right area of the main panel.

27

Example of setting up a Python environment:

<https://www.youtube.com/watch?v=AxSwTvnwCUU&t=45s>



Good luck

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